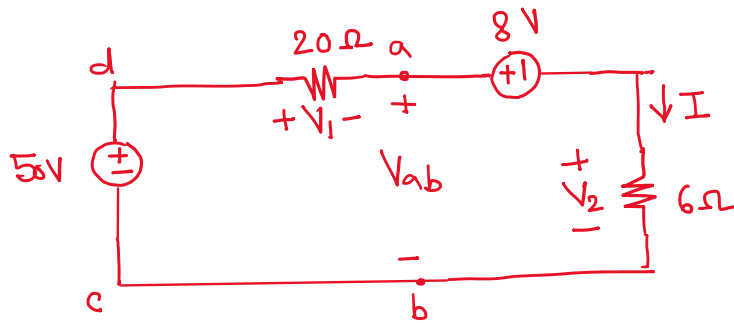


Voltage between two points

Wednesday, October 29, 2025 8:13 PM



$$+50 - V_1 - 8 - V_2 = 0$$

$$\Rightarrow 50 - 20I - 8 - 6I = 0 \text{ [Applying Ohm's Law]}$$

$$\Rightarrow I = \frac{42}{26}$$

V_{ab} : Voltage between Node a & Node b
[between two nodes there is always voltage]

V_{ab} : a is in higher potential wrt b.

V_{ab} : Voltage between two nodes even there is no element.
since voltage can exist. c-d-a-b can be considered a closed loop where KVL can be applied.

$$+50 - 20I - V_{ab} = 0 \Rightarrow V_{ab} = 50 - 20I$$

$$\text{If } b \text{ is the ground } V_{ab} = V_a - V_b = V_a$$

Voltage is always measured wrt a reference. If the ref is not defined, voltage at any node cannot be assigned.
If ref is defined, voltage is measured with respect to ref.
If ref is ground, its voltage is zero